CLASS 8 · CBSE · GANITA PRAKASH PART 1 · FORMULA BANK

Chapter 2: Power Play

Every formula and result of the chapter on one sheet

1	Meaning	$n^a = n \times n \times \ldots \times n$ (a times)
2	Product, same base	$n^a \times n^b = n^{a+b}$
3	Quotient, same base	$n^a \div n^b = n^{a-b}, n \neq 0$
4	Power of a power	$(n^a)^b = n^{ab}$
5	Product, same exponent	$m^a \times n^a = (mn)^a$
6	Quotient, same exponent	$m^a \div n^a = \left(\frac{m}{n}\right)^a, n \neq 0$
7	Zero exponent	$n^0 = 1, n \neq 0$
8	Negative exponent	$n^{-a} = \frac{1}{n^a}, n \neq 0$
9	Negative base	Even power positive, odd power negative
10	Codes	k places with n symbols each give n^k codes
11	Scientific notation	$x \times 10^y$ with $1 \leq x < 10$, y an integer
12	Lakh, crore, arab	$10^5, 10^7, 10^9$
13	Million, billion, trillion	$10^6, 10^9, 10^{12}$